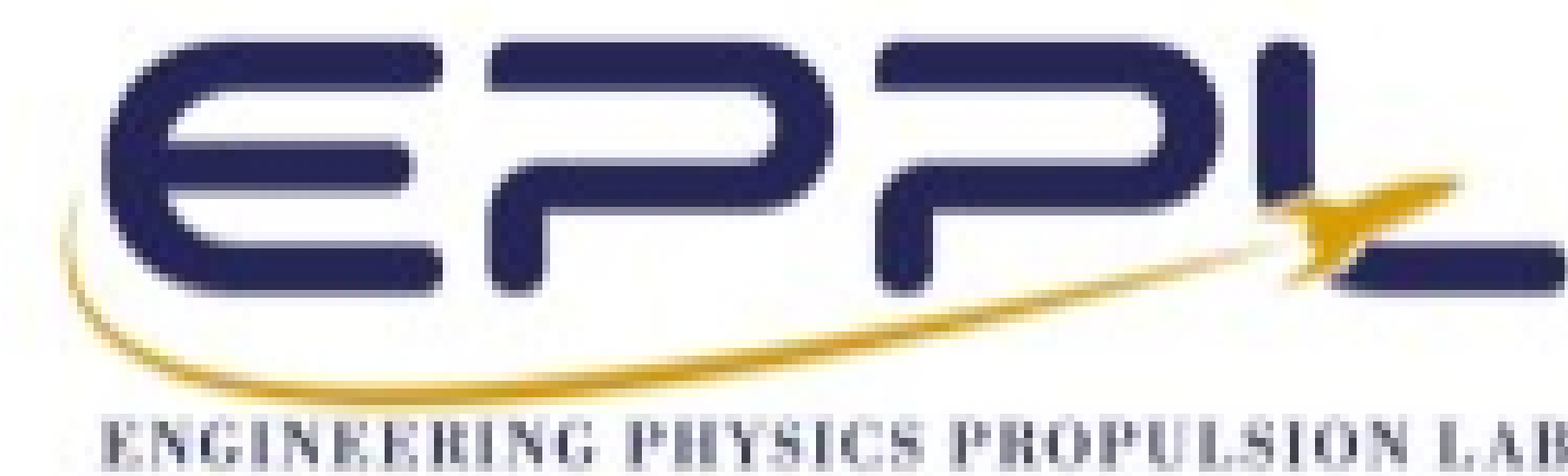




Robust Real-Time UAV Target Tracking with Onboard Vision-Based Yaw Controller



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Motivation

Challenge: Continuous Target Tracking

Autonomous UAV's require uninterrupted control commands to maintain stable flight and track moving targets. However, computer vision faces inherent challenges.

Detection Failures:

- Occlusion by environmental objects
- Motion blur during rapid movement
- Small target size at distance

Real Time Constraints

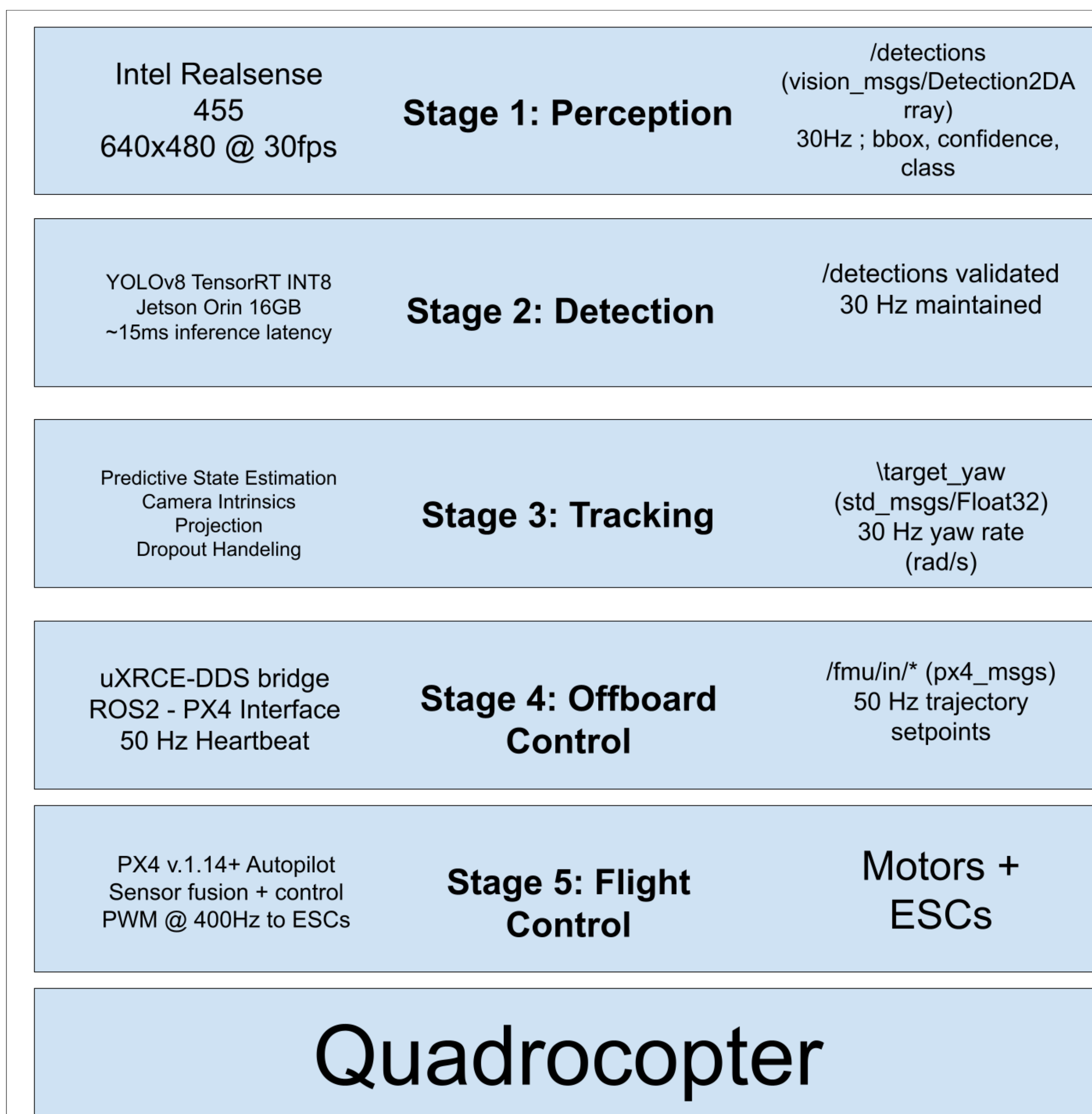
- PX4 autopilot requires >2Hz commands
- Target performance: 30 Hz for smooth tracking

Safety Critical

- Timeout triggers a failsafe mode
- Loss of target = mission failure
- Need graceful degradation

Solution: Predictive tracking maintains 30 Hz output during dropouts using motion estimation

Architecture



Hardware Platform

Vision:

- Intel RealSense D455
- Resolution: 640x480 (up to 1280x720)
- Frame Rate: 30 fps
- FOV: 87° x 58°

Compute:

- NVIDIA Jetson Orin 16GB

Autopilot:

- PX4 v1.14+ (Pixhawk Compatible)
- IMU: 6-axis gyro/accel

Software:

- ROS2 Humble Framework
- uXRCE-DDS Bridge Software

Predictive Tracking Algorithm

Challenge:

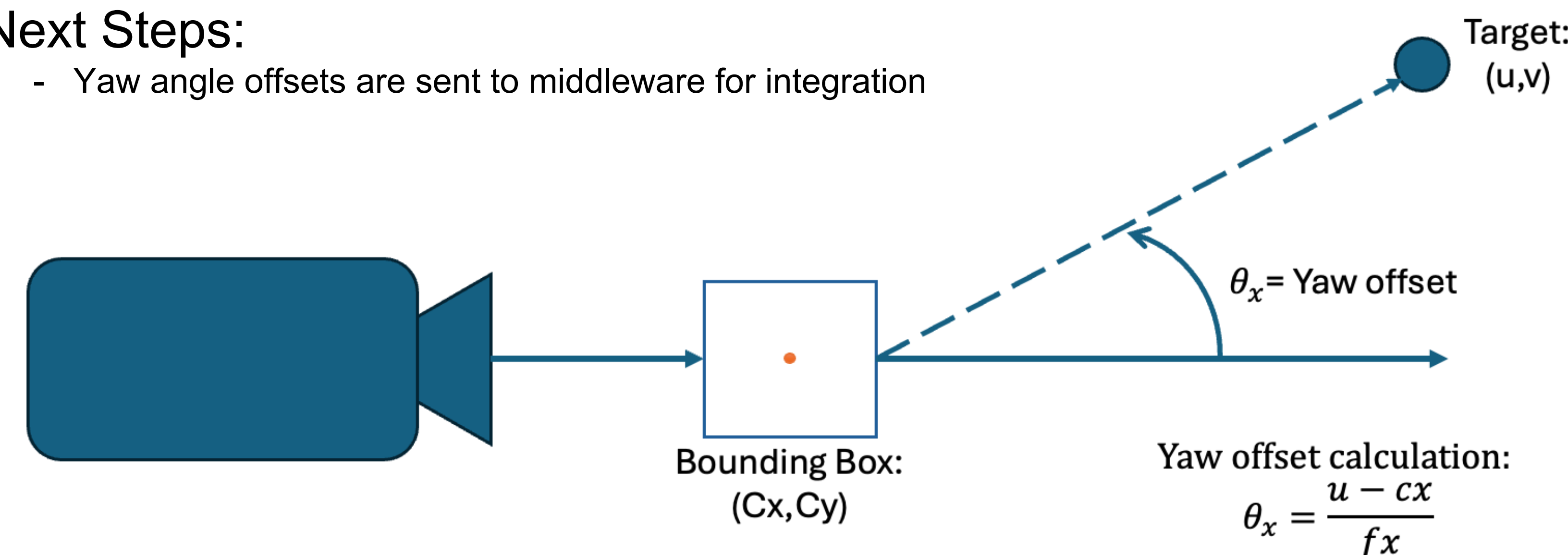
- Converting 2D detections into 3D angular data for yaw corrections
- Establish predictive yaw setpoint in the instance of target detection loss

Algorithm & Geometry:

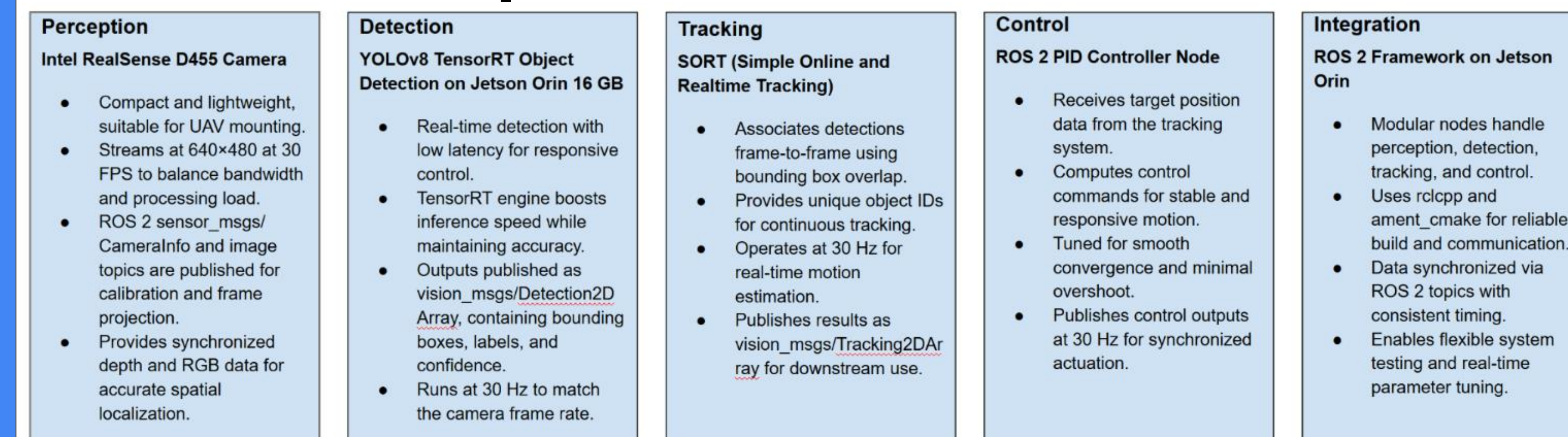
- Detection Input
 - Receives target's 2D bounding box coordinates relative to the camera's center
 - Each bounding box has a central coordinate (u,v) in pixel space with a specific confidence score
 - In the case of multiple targets, those with the highest confidence scores will be tracked
- Camera Intrinsics:
 - Camera's optical center (cx)
 - Camera's focal length (fx)
- Pinhole Camera Geometry:
 - Calculate yaw error through small-angle approximation
 - Convert horizontal displacement into angular offset (radians)
 - Positive = Right yaw offset, Negative = Left yaw offset
- 2D to 3D frame conversion:
 - Convert yaw offsets from camera's frame 2D to the UAV's 3D frame
 - Apply tf2 transforms through ROS2 for frame conversion
 - Apply Proportional (P) controller to ensure stable yaw correction motion
- Predictive yaw setpoint:
 - Use last computed yaw and yaw rate to determine next possible yaw offset angle
 - Apply linear extrapolation of previous yaw measurements

Next Steps:

- Yaw angle offsets are sent to middleware for integration



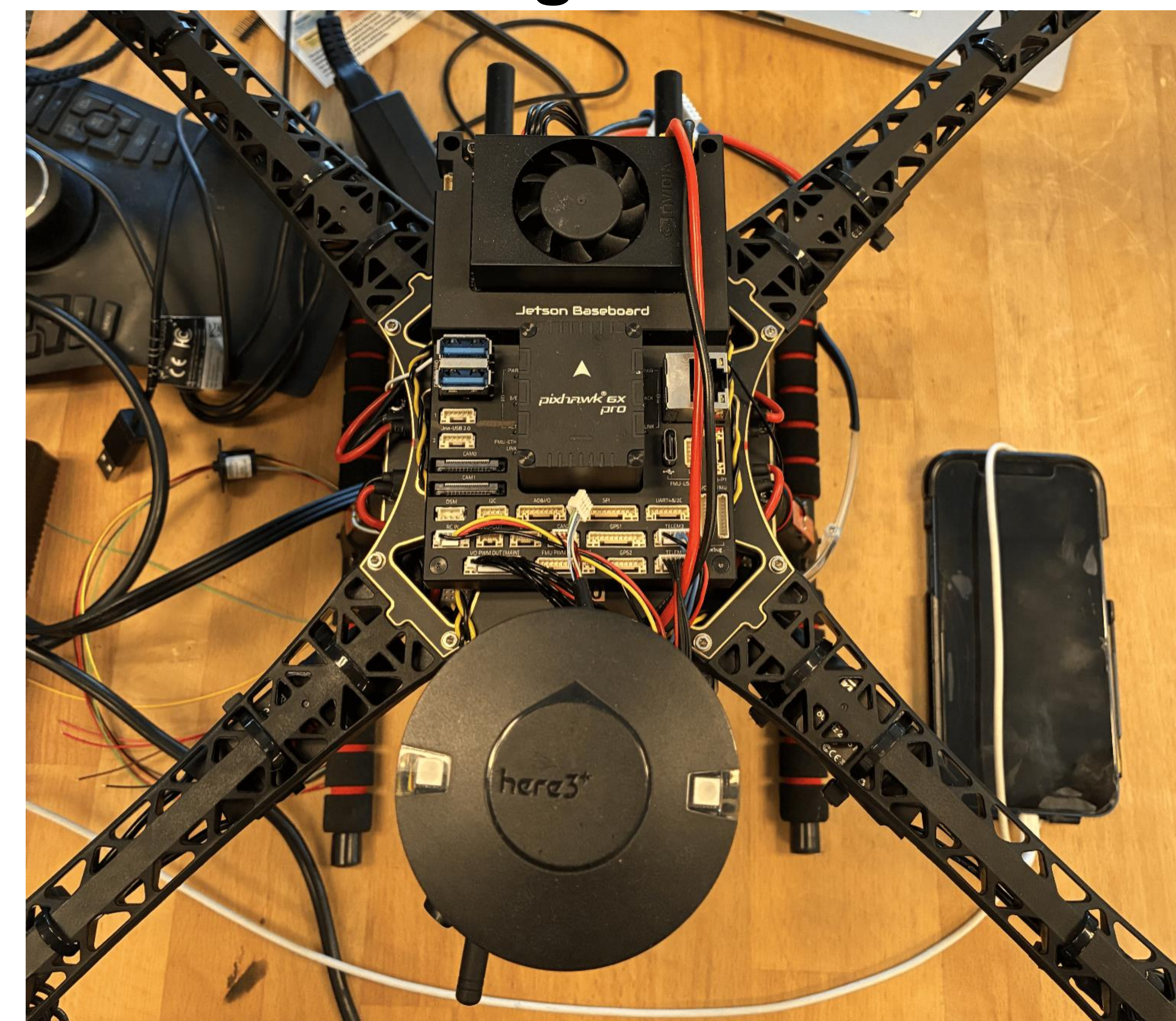
Implementation Details



Future Work

For the remainder of 2025, we will begin testing our hardware platform with the simulation framework you see here. This will give us a robust test platform to iterate on In Spring 2026, where we are targeting the SPIE Conference for difficult data in machine learning. For that, we will be revamping the computer vision architecture to use RF-DETR as opposed to the YOLOv8 CNN, as well as adding additional setpoint commands (potentially to 6 DoF, at least beyond current yaw tracking)

Hardware Integration



Full Flight Hardware Implementation: Jetson integrated into Pixhawk Jetson Baseboard Module along with 6X Pro FC Module. Test Flights to begin with full detection pipeline end of 2025, and continuing Spring 2026 as we prepare for conference submission

Validation

Steady State Analysis (Simulated of end to end pipeline)

Metric	Detection	Tracking	Status
Mean Rate (Hz)	30.00	30.00	clear
Message Drops	0	0	none

Key Findings

- Perfect Steady State: after initialization, simulated pipeline holds at exactly 30 Hz after initialization
- Synchronized Pipeline: Tracking publishes at same rate as detections
- Production Ready: Sustained performance clears hardware testing

Conclusion/Acknowledgements

Achievements

- Novel predictive tracking maintains control during dropouts
- 30 Hz real-time validated over 600-second test
- End-to-end camera to middleware integration
- Ready for flight testing with Jetson baseboard and FC integration

Impact: Enables reliable autonomous UAV tracking in real-world conditions where perfect detection is impossible

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